

SOFTWARE REQUIREMENTS SPECIFICATION

NIVAAS

Residential Architecture & Home Design Platform

Version 1.1 · Detailed Edition

Draft for Review

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Software Requirements Specification

NIVAAS — Residential Architecture & Home Design Platform

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines, in detail, the functional and non-functional requirements for **NIVAAS** (working title) — a residential architecture and home-design discovery-and-services platform for the Indian market. It is written to a level of detail sufficient for engineering, QA and design teams to plan, build and test the system without further clarification on scope. The platform’s **functional scope** is benchmarked against the public feature set of makemyhouse.com; its **information architecture, visual design, navigation, component system, copy, and underlying implementation** are original and independently engineered — no layout, asset, or content is reproduced from the reference site.

1.2 Intended Audience

- **Engineering** (frontend, backend, DevOps) — implementation reference.
- **QA** — test-case derivation from functional/non-functional requirements.
- **UI/UX** — constraint boundary for the design system (see companion UI_UX_DESIGN.md).
- **Product/Stakeholders** — scope sign-off document.

1.3 Scope

1.3.1 In Scope (v1)

- Public catalog of House Plans, 3D Elevations and Interior Designs with structured filtering and free-text search.
- Rich design detail pages with gallery, specifications, related content, and lead-capture CTAs.
- Guest and registered-user browsing, favouriting, and lead submission (enquiry + consultation request).
- Registered-user dashboard (saved designs, lead status, profile).
- Admin CMS: catalog management, taxonomy, services/packages, lead management, blog/content, curated SEO landing pages, media library, site settings, audit log.
- SEO architecture: SSR/SSG/ISR rendering, dynamic metadata, JSON-LD structured data, segmented sitemaps, canonicalization, controlled indexing of filter states.
- Performance architecture targeting Core Web Vitals “Good” thresholds despite an image-heavy catalog.

1.3.2 Out of Scope (v1) — Deferred / Explicitly Excluded

Item	Status	Notes
Payments / checkout	Excluded v1	Services are lead-based (enquiry/consultation), not transactional, in v1
Architect/professional marketplace login & bidding	Deferred v2	Requires a distinct role, portfolio model, matching logic
Real-time chat	Excluded v1	Lead capture is form + async notification based
Native mobile apps (iOS/Android)	Excluded v1	Responsive web only
Full multi-language i18n framework	Excluded v1	English content; Hindi terms used contextually, not a full i18n system
Dedicated search engine (Elasticsearch/Algolia/Meilisearch)	Deferred	v1 uses PostgreSQL FTS/trigram; revisit at scale
Post-enquiry project lifecycle tracking (Project/ProjectFile)	Deferred v1.1 (pending decision)	Schema defined now to avoid future migration pain; UI/flows not built in v1 unless confirmed

1.4 Definitions, Acronyms & Abbreviations

Term	Definition
Design	Umbrella entity for a House Plan, Elevation, or Interior listing; shares common fields (title, slug, media, SEO) with a type-specific detail table
BHK	Bedroom-Hall-Kitchen count; standard Indian residential real-estate sizing convention
Vastu	Traditional Indian architectural/spatial compliance system; modeled as a boolean/filterable attribute, not validated by the system itself
Facing	Cardinal/ordinal orientation of a plot/plan entrance (N, S, E, W, NE, NW, SE, SW)
RBAC	Role-Based Access Control

JWT	JSON Web Token, used for stateless access-token authentication
ISR	Incremental Static Regeneration — Next.js strategy to regenerate static pages on a schedule or on-demand without a full rebuild
SSR / SSG	Server-Side Rendering / Static Site Generation
CWV	Core Web Vitals — LCP, INP, CLS
LCP	Largest Contentful Paint (loading performance)
INP	Interaction to Next Paint (responsiveness)
CLS	Cumulative Layout Shift (visual stability)
FTS	Full-Text Search
CDN	Content Delivery Network
CMS	Content Management System (the admin panel, in this context)
Programmatic SEO	Systematically generated, individually unique landing pages targeting long-tail search queries (e.g. "30x50 East Facing House Plan")
Thin content	A page with insufficient unique content to justify indexing (SEO anti-pattern the system must avoid for arbitrary filter permutations)

1.5 References

Document	Content
PRD.md	Product goals, personas, user stories, success metrics
ARCHITECTURE.md	Full technology stack, folder structures, rendering/caching/search/security architecture
DATABASE_SCHEMA.md	Complete entity field tables and indexing notes
API_ROUTES.md	Full REST endpoint index with auth/role requirements
USER_FLOWS.md	Step-by-step guest/user/admin flow diagrams
UI_UX_DESIGN.md	Navigation, design tokens, page-by-page component breakdown
PROJECT_PLAN.md	Phase plan, testing strategy, deployment/environment detail

This SRS is the authoritative scope document; the companion documents above are the authoritative *implementation-detail* documents for their respective areas and are summarized/incorporated here where relevant to requirement traceability.

1.6 Document Conventions

- Functional requirements are identified FR-**<AREA>-<n>** and stated with “shall” (mandatory, testable).
- Non-functional requirements are identified NFR-**<AREA>-<n>**.
- Priority: **M** = Must have (v1 launch blocker), **S** = Should have (v1 target, not launch-blocking), **C** = Could have (v1.1+ candidate).

2. Overall Description

2.1 Product Perspective

NIVAAS is a new, greenfield, standalone web platform — it does not integrate with or depend on any existing internal system. It is composed of three logical tiers plus a media/edge layer:



Full technology table and diagram detail: ARCHITECTURE.md, Sections 1-2.

2.2 Product Functions (Summary)

1. Structured + free-text search and filtering across three design types.
2. Rich, SEO-optimized design detail presentation.
3. Lead capture (enquiry, consultation request) with spam mitigation.
4. Account system with favourites and lead tracking.
5. Admin CMS for full catalog, content, services, leads, taxonomy, SEO and settings management.
6. Automated SEO scaffolding: metadata, structured data, sitemaps, controlled indexing.
7. Performance-engineered image delivery pipeline.

2.3 User Classes and Characteristics

Class	Technical proficiency	Frequency of use	Primary goals
Guest	Low-medium (general public)	One-off to occasional	Find a house plan/elevation/interior matching plot and preferences; contact for services
Registered User	Low-medium	Occasional (revisits while planning a build, weeks-to-months horizon)	Save/compare designs, track enquiry/consultation status
Administrator	Medium-high (internal staff, trained on the CMS)	Daily	Publish and maintain catalog content, manage incoming leads, maintain SEO landing pages and site content

2.4 Operating Environment

- **Client:** modern evergreen browsers (Chrome, Safari, Firefox, Edge, last 2 major versions), responsive from ~360px (mobile) to large desktop (over 1920px).
- **Server:** containerized Node.js runtime (backend) and Next.js runtime (frontend), cloud-hosted; PostgreSQL managed instance; S3-compatible object storage; CDN in front of both static assets and cacheable pages.
- **Environments:** local, staging, production, each with isolated database and storage (see PROJECT_PLAN.md, “Deployment & Environment”).

2.5 Design and Implementation Constraints

- **C1:** Frontend must be a React-based framework with first-class SSR/SSG/ISR support – a plain client-rendered SPA is explicitly rejected because of the SEO requirement (programmatic landing pages must be crawlable without JS execution dependency).
- **C2:** Database must be PostgreSQL, accessed via Prisma ORM (team standard, strong TypeScript type-safety).
- **C3:** No content, imagery, layout or copy may be copied from the reference benchmark site; all catalog media must be original or properly licensed.
- **C4:** v1 search must not depend on an external search service; PostgreSQL native FTS/trigram is the required v1 approach.
- **C5:** All mutating API input must be validated with Zod before reaching business logic – no unvalidated writes.
- **C6:** Arbitrary filter-combination URLs must never be silently indexed – SEO safety is a hard constraint, not an optimization.

2.6 Assumptions and Dependencies

- **A1:** Original/licensed photography and plan imagery will be available ahead of catalog population; the system does not generate design imagery itself.
- **A2:** A specific object storage/CDN vendor will be chosen at infrastructure setup; this SRS defines required *capabilities* (variant generation, edge caching), not a vendor.
- **A3:** Admins will curate which filter combinations become indexable SEO landing pages; the system provides the mechanism but not the SEO content strategy itself.
- **A4:** Legal/compliance review (privacy policy, terms, data retention for lead PII) happens alongside development and is reflected in Section 5.7.

2.7 Apportioning of Requirements

Requirements marked C (Could-have) – dedicated search engine, professional marketplace role, project lifecycle UI, payments, native apps, full i18n – are explicitly deferred and may be apportioned to v1.1/v2 without altering the v1 data model where the schema has already been designed to accommodate them (e.g. Project/ProjectFile tables exist in the schema now to avoid a breaking migration later).

3. System Features (Functional Requirements)

3.1 Catalog Browsing & Search

Description: Visitors discover designs via structured filters, category browsing, or free-text search that resolves to structured filters and, where possible, a curated SEO landing page.

ID	Requirement	Priority
FR-CAT-1	The system shall allow filtering House Plans by plot width, plot depth, unit (ft/m), facing, BHK, floors, bathrooms, parking availability, Vastu compliance, plot type (regular/corner), style and budget range.	M
FR-CAT-2	The system shall allow filtering Elevations by floors, plot width, facing and architectural style.	M
FR-CAT-3	The system shall allow browsing Interiors by room type (living room, bedroom, kitchen, bathroom, pooja room, other) and style.	M
FR-CAT-4	The system shall support free-text search that parses plot dimensions, facing keywords and design-type keywords into structured filter parameters (e.g. “30x50 east facing duplex” maps to width=30, depth=50, facing=EAST, styleTag~duplex).	M
FR-CAT-5	The system shall present a paginated, sortable (relevance / newest / most popular) grid of matching designs for any filter or search state.	M
FR-CAT-6	The system shall render curated, high-value filter combinations as statically generated, indexable landing pages with unique supporting copy.	M
FR-CAT-7	The system shall mark any non-curated (arbitrary/thin) filter combination as non-indexable (noindex) and shall exclude it from all sitemaps.	M
FR-CAT-8	The system shall allow combining multiple filters simultaneously and shall reflect the active filter state in the URL for shareability.	S
FR-CAT-9	The system shall display a “no results” state with suggested nearby filter relaxations when a filter combination returns zero designs.	S

3.2 Design Detail Presentation

Description: Every design (regardless of type) is presented through a consistent detail-page template with type-appropriate specifications.

ID	Requirement	Priority
FR-DET-1	The system shall display, for every design, an image gallery, a structured specification table appropriate to its type, a description, and a feature list.	M
FR-DET-2	The system shall display a “similar/related designs” section computed from shared category/style/attributes.	M
FR-DET-3	The system shall allow any authenticated user to save or unsave a design as a favourite.	M
FR-DET-4	When a guest attempts to save a favourite, the system shall prompt authentication and, upon success, shall automatically complete the originally intended save action.	S
FR-DET-5	The system shall provide “Request customization” and “Talk to an architect” calls-to-action on every design detail page, each leading into the enquiry flow (Section 3.4) pre-associated with that design.	M
FR-DET-6	The system shall render breadcrumbs, a canonical URL, OpenGraph metadata and JSON-LD structured data on every design detail page.	M
FR-DET-7	The system shall provide a share affordance	S

	(copy link / native share) on every design detail page.	
FR-DET-8	The system shall lazy-load below-the-fold gallery images and shall reserve fixed aspect-ratio space for all images to prevent layout shift.	M

3.3 Accounts & Authentication

ID	Requirement	Priority
FR-ACC-1	The system shall allow registration with name, email and password, with client- and server-side validation.	M
FR-ACC-2	The system shall allow login with email and password.	M
FR-ACC-3	The system shall issue a short-lived JWT access token and a rotating, httpOnly-cookie refresh token on successful authentication.	M
FR-ACC-4	The system shall allow a user to log out, which shall revoke the active refresh session server-side.	M
FR-ACC-5	The system shall allow a registered user to view and edit their profile (name, phone, avatar).	S
FR-ACC-6	The system shall allow a registered user to view their saved designs, with unsave available inline.	M
FR-ACC-7	The system shall allow a registered user to view their enquiries and consultation requests with current status.	M
FR-ACC-8	The system shall rate-limit authentication endpoints to mitigate credential-stuffing/brute-force attempts.	M

3.4 Services & Lead Capture

ID	Requirement	Priority
FR-SVC-1	The system shall list active architectural services (Floor Plan Design, 3D Elevation, Structural Drawing, Working Drawings, Electrical Drawing, Plumbing Drawing, Interior Design, Vastu Consultation, Custom House Design) with descriptions.	M
FR-SVC-2	The system shall present package options (name, starting price, feature list) per service where defined.	S
FR-SVC-3	The system shall provide a consultation-request form capturing plot width/depth, budget range, requirement details and preferred contact time.	M
FR-SVC-4	The system shall provide an enquiry form on design and service pages capturing name, email, phone and message, associated with the originating design or service where applicable.	M
FR-SVC-5	The system shall rate-limit and validate public enquiry/consultation submission endpoints to mitigate spam and abuse.	M
FR-SVC-6	The system shall allow administrators to view, filter (by status/date/design/service) and update the status of enquiries (NEW to CONTACTED to IN_PROGRESS to CLOSED).	M
FR-SVC-7	The system shall allow administrators to view, filter and update the status of consultation requests (NEW to SCHEDULED to COMPLETED / CANCELLED).	M
FR-SVC-8	The system shall allow administrators to attach internal notes to any enquiry or consultation request, visible only to Admins.	S

3.5 Admin CMS - Catalog Management

ID	Requirement	Priority
FR-ADM-1	The system shall allow administrators to create, edit, publish, unpublish and archive designs of any type (House Plan, Elevation, Interior), including type-specific attributes.	M
FR-ADM-2	The system shall allow administrators to upload media for a design and shall automatically generate optimized responsive image variants (thumbnail/medium/large, WebP/AVIF).	M
FR-ADM-3	The system shall auto-suggest a unique slug from the design title, editable by the admin prior to publish.	S
FR-ADM-4	The system shall prevent publishing a design with a duplicate slug.	M
FR-ADM-5	The system shall support Draft and Published states with a preview capability for Draft designs (admin-only access).	M
FR-ADM-6	The system shall trigger cache/ISR revalidation for affected listing and detail pages, and sitemap regeneration, upon publish or update of a design.	M

3.6 Admin CMS - Content & Configuration

ID	Requirement	Priority
FR-ADM-7	The system shall allow administrators to manage taxonomy: categories, styles, tags and amenities, each attachable to designs.	M
FR-ADM-8	The system shall allow administrators to manage blog posts (title, slug, content, category, SEO fields, publish status).	S
FR-ADM-9	The system shall allow administrators to manage FAQs and testimonials, optionally scoped to a specific design or service.	S
FR-ADM-10	The system shall allow administrators to create and manage curated SEO landing pages, each defined by a path, a filter configuration, unique supporting copy, and SEO metadata.	M
FR-ADM-11	The system shall allow administrators to manage global site settings (site name, contact info, social links, default SEO fields, feature flags) via a key-value settings interface.	S
FR-ADM-12	The system shall allow administrators to manage user accounts, including role assignment.	S
FR-ADM-13	The system shall record an audit-log entry (actor, action, entity, timestamp, metadata) for every administrative create, update, publish and delete action.	M
FR-ADM-14	The system shall present administrators with a dashboard summarizing key counts (published designs by type, open leads this week, recent activity).	C

3.7 SEO & Content Discoverability

ID	Requirement	Priority
FR-SEO-1	The system shall generate segmented XML sitemaps for designs, services and blog content, aggregated under a sitemap index.	M
FR-SEO-2	The system shall generate a robots.txt disallowing crawl of thin/arbitrary filter-combination URLs.	M
FR-SEO-3	The system shall emit JSON-LD structured data appropriate to page type: CreativeWork/Product-style for designs,	M

	BreadcrumbList on all indexable pages, FAQPage where FAQs are present, Organization on the home page.	
FR-SEO-4	The system shall set a canonical URL on every indexable page, resolving filter-state duplicates to their canonical curated landing page where one exists.	M
FR-SEO-5	The system shall regenerate affected sitemaps and trigger ISR revalidation automatically on content publish/update - no manual SEO ops step required for routine publishing.	S

4. External Interface Requirements

4.1 User Interfaces

Full specification (navigation structure, design tokens for light/dark themes, page-by-page component breakdown, responsive and accessibility requirements) is defined in **UI_UX_DESIGN.md**. Summary constraints binding on the UI: - **UI-1**: Primary navigation shall expose House Plans, Elevations, Interiors, Services and Inspiration as first-level items (not nested under a generic “Explore”). - **UI-2**: Listing pages shall present filters via a sidebar on desktop (md breakpoint and above) and a bottom-sheet drawer on mobile. - **UI-3**: The admin panel shall use a persistent grouped sidebar (Designs, Services, Enquiries, Consultations, Users, Taxonomy, Media Library, Blog, SEO Pages, FAQs, Testimonials, Site Settings, Audit Log). - **UI-4**: All images shall use descriptive alt text sourced from MediaAsset.altText, editable by admins.

4.2 API Interfaces

The frontend communicates with the backend exclusively via a versioned JSON REST API (/api/v1/...) with a consistent response envelope:

```
success: { "success": true, "data": { }, "meta": { "page": 1, "pageSize": 20, "total": 134 } }
error: { "success": false, "error": { "code": "VALIDATION_ERROR", "message": "...", "details": [ ] } }
```

The full endpoint index (method, path, required role, description) across Auth, Users, Designs, Taxonomy, Services, Enquiries/Consultations, Projects, Blog/Content, SEO, and Admin/Media/Settings/Audit resources is defined in **API_ROUTES.md** and is considered part of this SRS by reference.

4.3 Hardware Interfaces

None beyond standard end-user client devices (desktop, laptop, tablet, mobile) capable of running a modern browser; no specialized hardware is required.

4.4 Software Interfaces

Interface	Purpose
PostgreSQL (via Prisma ORM)	Primary relational data store for all entities
S3-compatible object storage	Storage of original media uploads and generated responsive variants
CDN	Edge caching/delivery of static assets, media variants, and cacheable SSG/ISR pages
Email/notification provider (TBD at infra setup)	Lead-notification delivery to admins on new enquiry/consultation submission

4.5 Communications Interfaces

All client-server communication occurs over HTTPS/TLS. The refresh token is transmitted exclusively via a secure, httpOnly, SameSite cookie; the access token is used as a Bearer header on API requests.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-PERF-1	Largest Contentful Paint (LCP) shall be <= 2.5s on catalog listing and detail page templates under typical network/device conditions (measured

	via Lighthouse CI on representative pages).
NFR-PERF-2	Interaction to Next Paint (INP) shall be $\leq 200\text{ms}$ on the same templates.
NFR-PERF-3	Cumulative Layout Shift (CLS) shall be ≤ 0.1 .
NFR-PERF-4	The system shall never serve an unoptimized original image directly to a client; all served images shall pass through the responsive AVIF/WebP variant pipeline.
NFR-PERF-5	Listing API responses shall return lightweight “card” DTOs (not full detail payloads) to minimize payload size.
NFR-PERF-6	The system shall apply route-based code splitting and lazy-load below-the-fold UI components on content-heavy pages.

5.2 Security

ID	Requirement
NFR-SEC-1	Passwords shall be stored using a strong one-way adaptive hash (bcrypt or argon2), never in plain text or reversible encryption.
NFR-SEC-2	All mutating API endpoints shall validate input against a Zod schema before any business logic executes.
NFR-SEC-3	Admin-only endpoints shall be protected by RBAC middleware at the API layer, independent of and in addition to any UI-level restriction.
NFR-SEC-4	File uploads shall be validated by MIME-type allowlist and maximum size, and shall be re-encoded server-side before storage (client-declared type shall not be trusted).
NFR-SEC-5	Public form endpoints (registration, login, enquiry, consultation) shall be rate-limited per IP/account.
NFR-SEC-6	The system shall use parameterized queries exclusively (via Prisma) - no raw string-interpolated SQL.
NFR-SEC-7	The system shall apply standard security headers (via Helmet or equivalent) and a strict CORS allowlist.
NFR-SEC-8	All administrative mutations shall be recorded in an immutable audit log.
NFR-SEC-9	Secrets (DB credentials, JWT signing keys, storage keys) shall be provided via environment variables or a secret manager, never committed to source control.

5.3 SEO & Discoverability

ID	Requirement
NFR-SEO-1	Every indexable page shall have a unique title, meta description, and canonical URL.
NFR-SEO-2	Arbitrary/thin filter-combination URLs shall be excluded from all sitemaps and marked noindex.
NFR-SEO-3	Sitemaps shall be regenerated (or incrementally updated) within a bounded time (target: within 15 minutes) of relevant content publish/update.

5.4 Reliability & Availability

ID	Requirement
NFR-REL-1	The system shall degrade gracefully if object storage/CDN is briefly unavailable (e.g. render a placeholder rather than failing the page).
NFR-REL-2	Database schema changes shall be applied through a controlled, reversible migration process (Prisma Migrate) as a distinct, auditable deployment step.
NFR-REL-3	The public catalog (browse/search/detail) shall target 99.5% monthly uptime at minimum.

5.5 Scalability

ID	Requirement
NFR-SCALE-1	The catalog query layer shall be indexed to remain performant up to at least 50,000 published designs without requiring an external search

	engine (v1 target).
NFR-SCALE-2	The architecture shall allow the search layer to be swapped for a dedicated search engine without changing the public API contract (service-layer abstraction around search).

5.6 Usability & Accessibility

ID	Requirement
NFR-USE-1	The UI shall meet WCAG 2.1 AA color-contrast requirements in both light and dark themes.
NFR-USE-2	All interactive elements shall be reachable and operable via keyboard, with a visible focus state.
NFR-USE-3	Forms shall present inline, field-level validation errors in plain language.

5.7 Legal, Privacy & Data Retention

ID	Requirement
NFR-LEGAL-1	The system shall publish a Privacy Policy and Terms of Service, linked from the footer and referenced at account registration and lead-form submission.
NFR-LEGAL-2	Personally identifiable information collected via enquiry/consultation forms (name, email, phone) shall be accessible only to authenticated Admin users.
NFR-LEGAL-3	The system shall support deletion of a user’s account and associated PII on request, subject to legitimate retention needs (e.g. audit log entries may retain a non-PII action record).

5.8 Maintainability

ID	Requirement
NFR-MAINT-1	All code shall pass ESLint and Prettier checks as a CI gate.
NFR-MAINT-2	Critical user and admin flows shall be covered by Vitest unit tests and Playwright end-to-end tests, executed in CI.
NFR-MAINT-3	Lighthouse CI shall enforce performance/SEO/accessibility budgets on key page templates as a CI gate.

6. Data Requirements

6.1 Conceptual Data Model

The data model centers on a shared **Design** parent entity (type: HOUSE_PLAN / ELEVATION / INTERIOR) with a 1:1 type-specific detail table, supported by: taxonomy (Category, Style, Amenity, Tag), media (MediaAsset, DesignMedia), accounts (User, RefreshSession, Favorite), commerce-adjacent leads (Service, ServicePackage, Enquiry, ConsultationRequest), post-lead lifecycle (Project, ProjectFile - schema present, UI deferred per Section 1.3.2), content (BlogPost, BlogCategory, FAQ, Testimonial), SEO (SeoLandingPage), configuration (SiteSetting), and governance (AuditLog). The complete field-level specification for every entity is defined in **DATABASE_SCHEMA.md** and is incorporated into this SRS by reference; a summary is provided below.

6.2 Entity Summary

Entity	Purpose	Key relationships
User	Account record	1-to-many RefreshSession, Favorite, Enquiry, ConsultationRequest, Project
Design	Shared parent for all catalog listings	many-to-1 Category, many-to-1 Style; 1-to-1 type detail table; 1-to-many DesignMedia; many-to-many Amenity/Tag
HousePlanDetails / ElevationDetails / InteriorDetails	Type-specific attributes	1-to-1 with Design
DesignMedia / MediaAsset	Gallery/floor-plan/cover imagery and generated variants	many-to-1 Design, referenced by other content entities
Category / Style / Amenity / Tag	Taxonomy for filtering and organization	many-to-many with Design via join tables

Favorite	User-saved designs	many-to-1 User, many-to-1 Design
Service / ServicePackage	Architectural service catalog	1-to-many ServicePackage, 1-to-many Enquiry
Enquiry / ConsultationRequest	Lead capture	many-to-1 User (nullable), many-to-1 Design/Service (nullable)
Project / ProjectFile	Post-lead project lifecycle (schema defined, v1.1 candidate)	many-to-1 User, 1-to-many ProjectFile
BlogPost / BlogCategory	SEO/content marketing	many-to-1 BlogCategory
SeoLandingPage	Curated indexable filter-combination pages	optional link to Category
FAQ / Testimonial	Supporting trust/content elements	optionally scoped to Design/Service
SiteSetting	Global key-value configuration	-
AuditLog	Immutable record of admin actions	many-to-1 User (actor)

6.3 Data Volume Assumptions

Initial catalog expected in the low thousands of designs; architecture and indexing (see NFR-SCALE-1) are specified to remain performant to at least 50,000 designs on the v1 PostgreSQL-only approach.

6.4 Data Retention

Lead PII (Enquiry, ConsultationRequest contact fields) is retained per NFR-LEGAL-2/3; AuditLog entries are retained indefinitely as a non-PII governance record even where the related user account is later deleted (actor reference may be anonymized rather than the log entry deleted).

7. Use Case Narratives

UC-1: Guest finds and enquires about a house plan

Actor: Guest. **Preconditions:** none. **Trigger:** visitor lands on Home or a search engine result. 1. Guest searches “30x50 east facing” or navigates House Plans and applies filters. 2. System returns a filtered/curated result set (FR-CAT-1 to FR-CAT-7). 3. Guest opens a design detail page (FR-DET-1 to FR-DET-6). 4. Guest selects “Talk to an architect” (FR-DET-5) and submits the enquiry form (FR-SVC-4). 5. System validates and rate-limit-checks the submission (FR-SVC-5), stores the Enquiry, and notifies Admins. **Postcondition:** Enquiry exists with status NEW, visible in Admin -> Enquiries.

UC-2: Registered user tracks a consultation request

Actor: Registered User. **Preconditions:** authenticated. 1. User browses Services, opens a Service detail page. 2. User submits the consultation-request form (FR-SVC-3), associated with their account. 3. Admin updates status NEW -> SCHEDULED -> COMPLETED over time (FR-SVC-7). 4. User checks Dashboard -> Consultation Requests at any point to see current status (FR-ACC-7). **Postcondition:** User has real-time visibility into lead status without contacting support.

UC-3: Admin publishes a new House Plan with SEO landing page

Actor: Administrator. **Preconditions:** authenticated as Admin. 1. Admin creates a new Design (type=HOUSE_PLAN), fills common + type-specific fields (FR-ADM-1). 2. Admin uploads gallery/floor-plan/cover media; system generates optimized variants (FR-ADM-2). 3. Admin reviews a Draft preview, then Publishes (FR-ADM-5); system validates slug uniqueness (FR-ADM-4) and triggers ISR revalidation + sitemap update (FR-ADM-6, FR-SEO-5). 4. Admin separately creates a curated SeoLandingPage referencing this plan’s filter signature (FR-ADM-10), writing unique supporting copy to avoid thin content. **Postcondition:** The design is publicly browsable/searchable and, where curated, has a dedicated indexable landing page.

UC-4: System resolves a free-text search to a canonical landing page

Actor: System (triggered by Guest/User query). 1. User enters “25x40 duplex house plan” in global search. 2. Query parser extracts structured filters (FR-CAT-4). 3. System checks for a matching curated SeoLandingPage.filterConfig. 4. If matched, user is routed to the canonical landing page (best SEO/UX outcome, FR-SEO-4); if not matched, a filtered /designs result set is rendered and marked noindex (FR-CAT-7).

8. Requirements Traceability (Phase Mapping)

Requirement group	Primary phase(s)	Companion doc
FR-ACC-* (Accounts)	P1	ARCHITECTURE.md Section 8, USER_FLOWS.md Section 2

FR-CAT-, <i>FR-DET</i> - (Catalog, Detail)	P2-P3	DATABASE_SCHEMA.md, USER_FLOWS.md Section 1
FR-SVC-* (Services & Leads)	P4	USER_FLOWS.md Sections 3-4
FR-ADM-* (Admin CMS)	P2, P4, P5	UI_UX_DESIGN.md Section 4, USER_FLOWS.md Sections 6-8
FR-SEO-*	P5	ARCHITECTURE.md Section 9
NFR-PERF-, <i>NFR-SCALE</i> -	P6	ARCHITECTURE.md Section 6
NFR-SEC-, <i>NFR-LEGAL</i> -	P1, P7	ARCHITECTURE.md Section 8
NFR-MAINT-*, testing	P7	PROJECT_PLAN.md “Testing & Edge-Case Strategy”

Full phase deliverables: PROJECT_PLAN.md.

9. Risks & Open Issues

Risk / Issue	Impact	Mitigation / Owner action needed
Final brand name/domain not yet decided	Blocks final copy, metadata, legal pages	Product decision required before P5 (Content+SEO) content lock
“Projects” lifecycle scope (v1 vs v1.1) undecided	Schema is ready either way; UI/flow build effort depends on the call	Product decision recommended before P4
Paid vs free consultation policy unclear	Affects ServicePackage pricing display and CTA copy	Product/business decision needed before P4
Thin-content SEO risk if filter combinations are over-generated	Could suppress organic ranking	Enforced via FR-CAT-7/NFR-SEO-2 as a hard constraint, admin-curated landing pages only
Image-heavy catalog risk to Core Web Vitals	Could miss LCP/CLS targets at scale	Mitigated by mandatory responsive pipeline (NFR-PERF-4) and CI-enforced Lighthouse budgets (NFR-MAINT-3)
No dedicated search engine in v1	Search quality/scale ceiling at very large catalog sizes	Architecture keeps search behind a service-layer abstraction (NFR-SCALE-2) for future swap-in

Appendix A - Development Phases

Phase	Name	Focus
P0	Foundation	Monorepo, TS strict, DB/ORM init, env, lint, error format, logging
P1	Auth + RBAC	Registration, login, refresh, RBAC middleware
P2	Core Catalog	Design model + type details, taxonomy, admin CRUD, media pipeline
P3	Discovery	Listing, filters, search parsing, favourites, related designs
P4	Services & Leads	Services/packages, enquiry/consultation forms, lead management
P5	Content + SEO	Blog, FAQs, curated SEO pages, metadata, JSON-LD, sitemaps, robots
P6	Performance	CDN/image pipeline, caching/ISR tuning, index audit, bundle optimization
P7	QA & Hardening	Unit/E2E coverage, RBAC tests, SEO crawl validation, accessibility, load/security audit

Full detail: PROJECT_PLAN.md.

Appendix B - Glossary

See Section 1.4. Additional CMS-specific terminology is defined inline in DATABASE_SCHEMA.md and UI_UX_DESIGN.md.

Appendix C - Companion Documents

Appendix D - Revision History

Version	Date	Change
1.0	September 2026	Initial SRS draft
1.1	September 2026	Expanded to full detailed edition: use cases, traceability matrix, risk register, legal/retention requirements, priority tagging on all functional requirements